



Answering from the Data

XPAND — AI & Automation Officer Assessment · Part 2

Product Availability Agent

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Contents

1. Problem	2
2. Solution	3
3. High-Level Workflow.....	3
4. Data Understanding.....	3
5. Questions the Agent Should Answer	4
6. Agent Behavior	4
7. Query Tool — query_availability	4
7.1 Tool Inputs.....	5
7.2 Example	5
8. Technical Architecture	5
9. AWS Architecture	6
10. Design Decisions	6
11. Alternative Approach Considered.....	7
12. Implementation & Testing.....	7
12.1 Deployed System.....	7
12.2 Dashboard.....	7
12.3 Availability Assistant and Arabic Support	8
12.4 Charts in the Assistant.....	8
12.5 Testing	8
13. Expected Outcome.....	9

1. Problem

The marketing team receives repetitive questions through the company's social channels, such as where a product can be purchased, which shop or area carries it, and whether it is available. The underlying POS availability data already reaches marketing from the ERP system. The current process requires someone to manually search the sheet for each question, which becomes slower as the dataset grows.

The assessment places the ERP-to-marketing data pipeline out of scope. The solution therefore focuses on making the supplied availability data easier for marketing to query and use.

2. Solution

The solution is an Availability Agent that accepts a natural-language marketing request, identifies the relevant filters, retrieves matching records through a deterministic query tool, and produces a concise answer grounded only in the returned POS data.

The solution is agent-centric rather than interface-centric. Marketing uses it through an authenticated internal dashboard deployed at xpanse.medgan.ai, in English or Arabic, but the core capability remains the agent plus the deterministic availability query.

3. High-Level Workflow

Part 2 – High-Level Workflow

Answering product availability questions from POS data

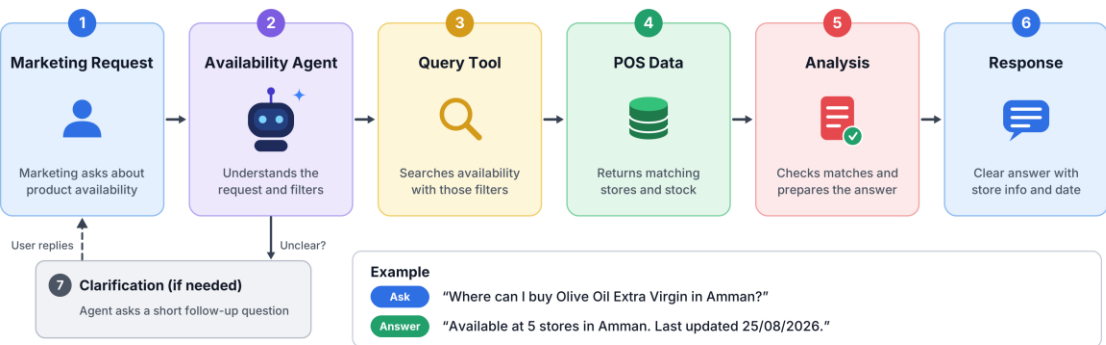


Figure 1 — High-Level Workflow

4. Data Understanding

The supplied sample POS availability dataset contains:

Attribute	Value
Records	936
Stores	40
Products	27

Cities	12
Categories	8
Availability states	In Stock, Low Stock, Out of Stock
Missing values	0

The dataset includes store, store type, area, city, region, product, category, pack size, quantity on shelf, availability status, shelf price, and last-updated information. Because availability is based on recorded POS observations rather than a guaranteed real-time inventory feed, responses should be framed as the latest recorded POS data and surface the last-updated timestamp when relevant.

5. Questions the Agent Should Answer

Capability	Example
Product availability	"Is Olive Oil Extra Virgin available?"
Where to buy	"Where can I buy Basmati Rice?"
City	"Where can I buy Basmati Rice in Amman?"
Area	"Which shops in Abdoun have Instant Coffee?"
Specific store	"Does Sameh Mall Abdoun have Basmati Rice?"
Pack size	"Where can I find 3 L Sunflower Cooking Oil?"
In-stock locations	"Which stores have Sunflower Cooking Oil in stock?"
Low-stock locations	"Which stores have low stock of Olive Oil?"

6. Agent Behavior

- Never invent store, product, price, quantity, or availability information.
- Extract the most specific filters available: product, pack size, city, area, store, and availability.
- Ask a clarification question when the request is ambiguous or lacks information required to identify the intended product or variant.
- Do not ask unnecessary questions when the request already contains enough information to query the data.
- Treat In Stock, Low Stock, and Out of Stock as distinct states.
- If no records match, clearly report that no matching records were found rather than fabricating an answer.
- Do not silently broaden a search. If a requested location has no match, explain this and ask whether the user wants a broader search.
- Represent availability as the latest recorded POS data and use the last-updated field to communicate freshness where relevant.

7. Query Tool — query_availability

The agent does not perform the data lookup through free-form natural-language reasoning. Instead, it calls a deterministic query tool with structured filters. This makes retrieval predictable and keeps the source of each answer explicit.

7.1 Tool Inputs

- **product_name** (optional)
- **pack_size** (optional)
- **city** (optional)
- **area** (optional)
- **store_name** (optional)
- **availability_status** (optional)

7.2 Example

Request: “Where can I find 3 L Sunflower Cooking Oil in Amman?”

Structured query: product_name="Sunflower Cooking Oil", pack_size="3 L", city="Amman"

The tool returns matching records containing the relevant store, area, city, product, pack size, availability, quantity, price, and last-updated information. The agent then turns those records into the final marketing-facing answer.

8. Technical Architecture

The technical design separates natural-language interpretation from deterministic retrieval. The Strands agent uses a Bedrock foundation model to understand and reason over the request, while query_availability performs the structured lookup against the POS dataset.

Part 2 – Technical Architecture

AI agent for product availability on AWS with Strands

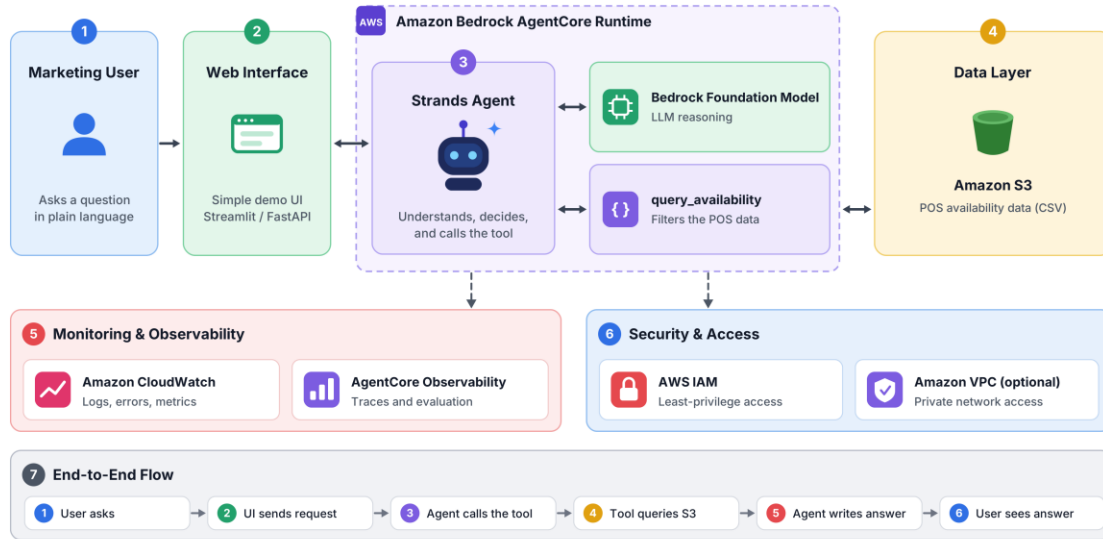


Figure 2 — Technical Architecture

9. AWS Architecture

The AWS design uses Amazon Bedrock AgentCore Runtime to host the Strands Availability Agent, Amazon Bedrock for the foundation model, Amazon S3 for the sample POS availability dataset, and a deterministic query tool for structured retrieval. AgentCore Observability and Amazon CloudWatch provide operational visibility, while IAM enforces least-privilege access. Marketing reaches the agent through an internal dashboard hosted on AWS Amplify, with users authenticated by Amazon Cognito and requests routed through Amazon API Gateway to the agent on AgentCore Runtime.

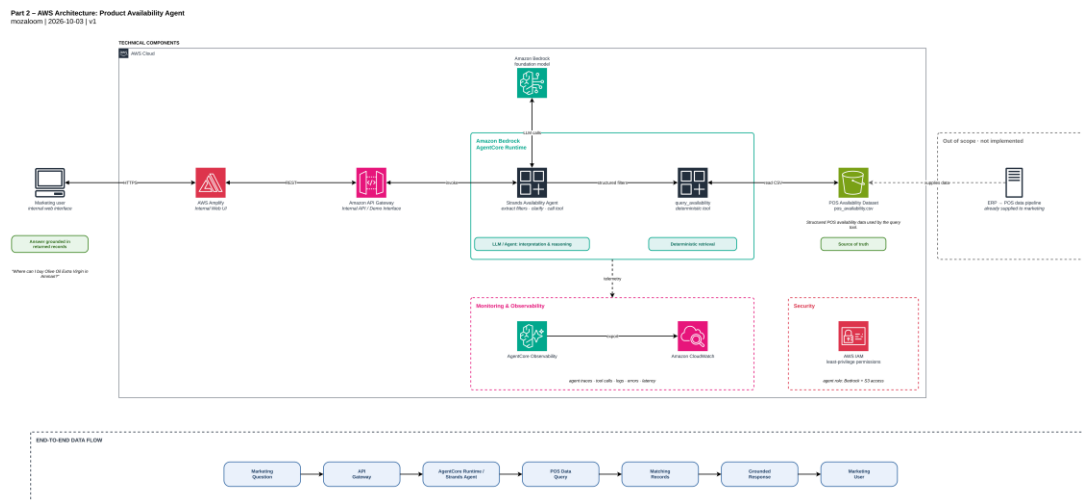


Figure 3 — AWS Architecture

10. Design Decisions

Decision	Rationale
Structured query instead of RAG/vector search	The supplied data is structured tabular availability data. Deterministic filtering is direct and easier to validate than embedding-based retrieval for this use case.
Agent + deterministic tool	The agent handles natural-language interpretation and clarification, while the tool performs the authoritative lookup. This reduces unsupported availability claims.
ERP pipeline out of scope	The assessment states that the data already reaches marketing automatically, so the solution does not redesign the upstream ERP pipeline.
Interface kept secondary	The authenticated internal dashboard delivers the capability to marketing, but the business solution is the availability agent and retrieval layer rather than a chatbot requirement.

11. Alternative Approach Considered

A traditional keyword/rule-based search interface was considered. It could work for a small, fixed set of phrases, but would require increasingly many rules as users phrase questions differently or omit filters. The chosen approach keeps deterministic retrieval while using an agent where natural-language interpretation and clarification provide value.

12. Implementation & Testing

The Product Availability Agent is implemented and deployed as an authenticated internal web application at xpanse.medgan.ai. It answers marketing availability questions from the supplied POS data through the Strands Availability Agent and the deterministic query_availability tool, as described in Sections 6 to 9.

12.1 Deployed System

- **Front end:** an internal dashboard and Availability Assistant hosted on AWS Amplify.
- **Authentication:** Amazon Cognito restricts access to authorized internal users.
- **API layer:** Amazon API Gateway receives assistant requests from the dashboard and invokes the agent.
- **Agent:** the Strands Availability Agent runs on Amazon Bedrock AgentCore Runtime and uses a Bedrock foundation model to interpret each request.
- **Retrieval:** the agent calls the deterministic query_availability tool, which filters the POS availability data and returns the matching records.
- **Language:** the dashboard and the Availability Assistant support both English and Arabic, with right-to-left (RTL) layout when Arabic is selected.

12.2 Dashboard

The dashboard is the internal home screen. It shows the latest recorded POS availability across 40 stores, 27 products, and 12 cities, so marketing can see the overall picture before asking a specific question.

- In Stock, Low Stock, and Out of Stock totals.
- An availability-state distribution bar.
- Data freshness, based on the last-updated field.
- Ranked city and category bars.
- Products to Watch and Stores to Watch lists.
- Filters by city, category, and availability state, persisted in the URL so a filtered view can be shared or reopened.
- Clickable products and stores that open the exact underlying records in a side panel.
- Sorting, search, and full-screen views for detailed inspection.

12.3 Availability Assistant and Arabic Support

The Availability Assistant accepts natural-language questions in English or Arabic. Arabic questions are interpreted into the same structured query_availability filters as English questions, and the interface switches to a right-to-left layout for Arabic. Representative supported questions:

English	Arabic
"Is Olive Oil Extra Virgin available?"	«هل زيت الزيتون البكر الممتاز متوفر؟»
"Where can I buy Basmati Rice in Amman?"	«أين يمكنني شراء أرز بسمتي في عمان؟»
"Which shops in Abdoun have Instant Coffee?"	«أي المحلات في عبدون تتوفر فيها القهوة سريعة التحضير؟»
"Which stores have low stock of Olive Oil?"	«أي المتاجر لديها مخزون منخفض من زيت الزيتون؟»

12.4 Charts in the Assistant

Each assistant answer includes the matching records returned by query_availability, which can be viewed as a table or as a chart.

- **Table:** collapsible, sortable, searchable, and available in full screen.
- **Grouping:** by city, area, store, product, or stock state.
- **Measures:** record count, quantity on shelf, average price, and lowest price.
- **Automatic charts:** a chart is generated when the user asks for one, in English or Arabic.
- **Grounding:** charts are built only from the records returned for that answer, never from data outside the query result.

Example chart requests: "Show a chart of Sunflower Cooking Oil availability by city."

«اعرض رسماً بيانياً لتوفر زيت دوار الشمس حسب المدينة»

12.5 Testing

Representative scenarios were tested against the deployed system to cover the agent behavior in Section 6, the dashboard, and the assistant charts:

Scenario	Representative test	Behavior checked
Product lookup	"Is Olive Oil Extra Virgin available?"	Returns matching POS records and answers only from them.
City / area / store filtering	"Which shops in Abdoun have Instant Coffee?"	Applies the most specific location filter extracted from the request.
Pack-size filtering	"Where can I find 3 L Sunflower Cooking Oil?"	Passes pack_size to query_availability and returns only that variant.
Availability states	"Which stores have low stock of Olive Oil?"	Treats In Stock, Low Stock, and Out of Stock as distinct states.
Ambiguous requests	"Is oil available?"	Asks a clarification question instead of guessing the product or variant.
No-result requests	Product requested in a location with no matching records	Reports that no matching records were found and does not broaden the search silently.
Freshness handling	Any availability answer	Frames results as the latest recorded POS data and shows the last-updated date.
Arabic queries	«أين يمكنني شراء أرز بسمتي في عمان؟»	Extracts the same filters as the English question; interface renders right to left.
Dashboard filtering	City, category, and state filters	Totals, bars, and lists update; filters persist in the URL.
Chart generation	English and Arabic chart requests	Chart is generated from the records returned for that answer only.

13. Expected Outcome

Marketing can now submit a natural-language availability question, in English or Arabic, through the authenticated dashboard at xpanse.medgan.ai and receive a concise answer grounded in the latest recorded POS data, with the matching records available as a table or chart. The dashboard gives an at-a-glance view of availability across all stores, products, and cities, so the team no longer needs to search the POS availability sheet by hand.